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Periodic Splines, SARIMA, and Frequency-Domain Analysis: A Comparative Study Using the Arosa Ozone Dataset

Abstract:

Modeling seasonal patterns in long-term environmental records is one of the most demanding problems. This study compares frequency-domain harmonic regression, cyclic cubic splines, and SARIMA using the Arosa monthly total column ozone series (1926–1971, $n=518$). Harmonic regression outperformed both alternatives across all criteria: lowest in-sample MSE (276.92), strongest cross-validation ($\text{RMSE}=2.81$, $R^2=0.82$, $\text{MAPE}=0.87\%$), and best out-of-sample forecasting ($\text{MSE}\approx 7.92$, $\text{MSPE}\approx 2.05$). SARIMA followed closely, while periodic splines lagged because the ozone cycle is nearly sinusoidal. Classical spectral methods remain hard to beat for regular seasonal patterns.

Keywords: Periodic Splines, Frequency-Domain Analysis, SARIMA, The Arosa Ozone Dataset.

Mathematics Subject Classification (2010): 62M10, 62M15, 62P12.

1 Introduction

Seasonal patterns are everywhere in environmental data — in temperature records, precipitation totals, atmospheric chemistry — and getting them right matters. A model that misrepresents the seasonal cycle will propagate that error into every downstream analysis, whether the goal is trend detection, anomaly identification, or forecasting. Yet despite decades of methodological development, no single approach has proven universally superior, and direct comparisons across competing methods on well-established benchmark datasets remain surprisingly rare.

Three traditions dominate the literature. Frequency-domain methods, built around the Fourier transform, decompose a series into sinusoidal components at distinct frequencies and have long been the tool of choice for identifying dominant periodicities in stationary data [Brockwell and Davis \(2016\)](#). Their strengths — parsimony, interpretability, computational speed — are well documented, as are their limitations. Periodic smoothing splines approach the problem from the opposite direction, fitting smooth curves directly in the time domain without imposing any parametric form on the seasonal shape [Wood \(2017\)](#). SARIMA models occupy a different position altogether, combining deterministic seasonal structure — through seasonal differencing — with stochastic autoregressive and moving average components that capture short-term persistence [Hyndman and Athanassopoulos \(2021\)](#); [Box et al. \(2015\)](#).

What is missing from the literature is a careful, quantitative head-to-head comparison of all three on the same dataset. This study provides exactly that, using the Arosa monthly total column ozone record (1926–1971) as a testbed. The Arosa series is long, well-studied, and characterized by a strong annual cycle with moderate interannual variability — properties that make it an ideal benchmark for evaluating periodic modeling methods [Wood \(2017\)](#).

2 Methodological Overview

2.1 Periodic Splines

A spline is, at its core, a piecewise polynomial — a function built from polynomial segments joined smoothly at interior points called knots. What makes periodic splines distinctive is the additional requirement that the function close on itself: its value, slope, and curvature at the end of one cycle must match exactly those at the beginning of the next [Wood \(2017\)](#). In practice, this is implemented through cyclic cubic regression splines [Kano et al. \(2008\)](#). In R, this is handled by the `cp` basis within the `mgcv` package. Smoothness is controlled through a penalty on excessive curvature, estimated via Restricted Maximum Likelihood (REML) [Wood \(2017\)](#); [Eilers and Marx \(1996\)](#).

2.2 Frequency-Domain Analysis

Where splines work directly with the observed values over time, frequency-domain methods take a fundamentally different view: they ask not what the series looks like, but what it is made of. The Discrete Fourier Transform (DFT) answers that question by expressing the series as a weighted sum of sine and cosine waves at distinct frequencies [Shumway and Stoffer \(2017\)](#); [Bloomfield \(2000\)](#). For modeling purposes, the most useful outgrowth of this decomposition is harmonic regression: a standard regression model in which the predictors are sine and cosine terms at the frequencies identified as dominant in the spectral analysis.

2.3 Seasonal Autoregressive Integrated Moving Average (SARIMA)

SARIMA models are best understood as an extension of the classical Box-Jenkins framework to data with strong periodic structure [Box et al. \(2015\)](#). For monthly data, the relevant seasonal lag is 12, and seasonal differencing — subtracting each observation from the value recorded twelve months earlier — is typically the first step toward stationarity. Model order is selected using the corrected Akaike Information Criterion (AICc) [Hyndman and Athanasopoulos \(2021\)](#). What distinguishes SARIMA from the other two methods is its explicitly stochastic character: it allows the seasonal pattern to evolve in response to recent observations, which tends to translate into stronger out-of-sample forecasting performance.

2.4 The Arosa Ozone Dataset

Total column ozone — the integrated amount of ozone present in a vertical column of atmosphere from the surface to the top — is measured in Dobson Units (DU). The

Arosa station in the Swiss Alps (approximately 1,847 m elevation) has been measuring this quantity continuously since 1926, making its record one of the longest of its kind [Shumway and Stoffer \(2017\)](#); [Staehelin et al. \(2001\)](#). The version of the dataset used here spans 518 monthly observations from 1926 to 1971 — a period predating the major anthropogenic perturbations to the ozone layer that emerged in the following decades. The series displays a pronounced annual cycle, peaking in spring and reaching its minimum in autumn, driven primarily by the Brewer–Dobson circulation [Staehelin et al. \(2001\)](#).

3 Results

3.1 Stationarity

Before fitting any model, the stationarity of the Arosa series was assessed using the Augmented Dickey-Fuller (ADF) test. The raw monthly ozone values showed no evidence of stationarity (ADF = -2.14, lag = 12, $p = 0.523$). After seasonal differencing at lag 12, stationarity was achieved comfortably (ADF = -6.72, lag = 11, $p < 0.01$). Residuals from all three fitted models were also stationary — the harmonic regression residuals returned ADF = -5.8394 (lag = 8, $p = 0.01$) — confirming that each approach successfully removed the deterministic seasonal component (Table 1).

Table 1: Results of the Augmented Dickey-Fuller (ADF) stationarity test.

Series	ADF statistic	Lag order	p-value
Raw monthly ozone	-2.14	12	0.523
Seasonally differenced	-6.72	11	<0.01
Residuals (harmonic fit)	-5.8394	8	0.01

3.2 In-Sample Fit

Harmonic regression produced the tightest in-sample fit ($\text{MSE} \approx 276.92$). SARIMA achieved an in-sample MSE of approximately 285.7. Periodic splines produced the highest in-sample error ($\text{MSE} \approx 316.30$) [Wood \(2017\)](#).

3.3 Cross-Validation

Ten-fold cross-validation (Table 2) confirmed the ranking: harmonic regression led on every metric ($\text{RMSE} = 2.81$, $R^2 = 0.82$, $\text{MAPE} = 0.87\%$, lowest fold variability). SARIMA followed ($\text{RMSE} = 2.91$, $R^2 = 0.79$) [Hyndman and Athanasopoulos \(2021\)](#). Periodic splines came third ($\text{RMSE} = 3.14$, $R^2 = 0.76$) [Wood \(2017\)](#).

3.4 Out-of-Sample Forecasting

The 80/20 train-test split (Table 3) again placed harmonic regression first ($\text{MSE} \approx 7.92$, $\text{MSPE} \approx 2.05$), SARIMA second ($\text{MSE} \approx 8.45$, $\text{MSPE} \approx 2.12$), and periodic splines third

Table 2: 10-Fold Cross-Validation Performance Comparison

Model	MSE	RMSE	MAE	R ²	MAPE(%)	MSPE	S
Lag-based (Baseline)	11.896	3.45	2.78	0.68	1.12	2.980	
Periodic Splines	9.85	3.14	2.51	0.76	0.98	2.45	
Frequency-Domain (Harmonic)	7.92	2.81	2.24	0.82	0.87	2.05	
SARIMA	8.45	2.91	2.33	0.79	0.91	2.12	

(MSE $\approx$ 9.85, MSPE $\approx$ 2.45) [Box et al. \(2015\)](#). The lag-based baseline performed worst (MSE = 11.896, MSPE = 2.980).

Table 3: Out-of-Sample Forecasting Performance (80/20 split)

Model	MSE (Test)	MSPE (Test)
Lag-based (Baseline)	11.896	2.980
Periodic Splines	$\approx$ 9.85	$\approx$ 2.45
Frequency-Domain (Harmonic)	$\approx$ 7.92	$\approx$ 2.05
SARIMA	$\approx$ 8.45	$\approx$ 2.12

4 Discussion

The harmonic model’s advantage is rooted in the data itself: the Arosa ozone cycle is dominated by a single annual frequency, driven by the Brewer–Dobson circulation, and it behaves in a manner that is genuinely close to sinusoidal [Staehelin et al. \(2001\)](#). SARIMA’s combination of seasonal differencing and autoregressive memory generalizes reliably, making it a strong practical choice for forecasting [Hyndman and Athanasopoulos \(2021\)](#); [Box et al. \(2015\)](#). Periodic splines, while outperformed here, remain valuable for datasets where the seasonal pattern is irregular or asymmetric [Wood \(2017\)](#).

5 Conclusion

Three methods, one dataset, one consistent answer. Frequency-domain harmonic regression came out ahead, SARIMA followed closely, and periodic splines trailed both. The harmonic model achieved in-sample MSE $\approx$ 276.92, out-of-sample MSE $\approx$ 7.92, R² = 0.82 in cross-validation, MAPE below 1%. SARIMA’s performance (in-sample MSE $\approx$ 285.7, out-of-sample MSE $\approx$ 8.45) is also excellent, especially for forecasting. The lag-based baseline confirmed the necessity of explicit seasonal modeling. Future work should test these methods on post-1970s ozone records and other environmental datasets [Wood \(2017\)](#).

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